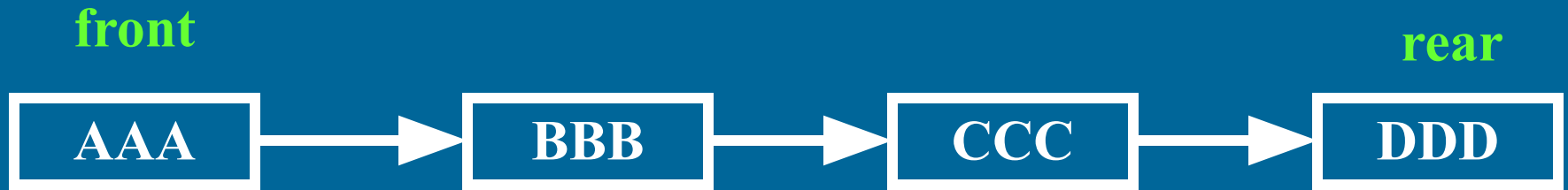


QUEUE

A queue is a linear list of elements in which deletions can take place only at one end, called the front, and insertions can take place only at the other end, called the rear.

The term “front” and “rear” are used in describing a linear list only when it is implemented as a queue.

Queues are also called **first-in-first-out** list



Suppose an element is deleted from the queue then it must be AAA



Suppose EEE is added to the queue and then FFF is added to the queue.



REPRESENTATIONS OF QUEUES

Each of our queues will be maintained by a linear array **QUEUE** and two pointer variables: **FRONT**, containing the location of the front element of the queue; and **REAR**, containing the location of the rear element of the queue.

The condition **FRONT = NULL** will indicate that the queue is empty.

Observe that whenever an element is deleted from the queue, the value of FRONT is increased by 1.

FRONT := FRONT + 1

Observe that whenever an element is added to the queue, the value of REAR is increased by 1

REAR := REAR + 1

PRACTICE

FIGURE 6-16

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PRACTICE

EXAMPLE 6.10

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Procedure-1

QINSERT(Queue, N, FRONT, REAR, ITEM)

Inserting an element into a queue

1. [queue already filled]
if $FRONT = 1$ and $REAR = N$, then:
 Write OVERFLOW, and Return
2. [find new value of REAR]
if $FRONT = NULL$, then: [queue initially empty]
 set $FRONT := 1$ and $REAR := 1$
else
 set $REAR := REAR + 1$;
[end of IF structure]
3. Set $queue[REAR] := ITEM$
4. return

Procedure-2

QDELETE(QUEUE, N, FRONT, REAR, ITEM)

Deletion of an element from a queue

1. [queue already empty]
if FRONT = NULL, then: Write UNDERFLOW, and Return
2. Set ITEM := QUEUE[FRONT]
3. [find new value of FRONT]
if FRONT = REAR then: [queue has only one element to start]
 set FRONT := NULL and REAR := NULL
else
 set FRONT := FRONT + 1;
[end of IF structure]
4. return

DEQUES

A deque is a linear list in which elements can be added or removed at either end but not in the middle.

The term deque is a contraction of the name double-ended queue

DEQUES...(cont)

There are two variations of the deque:

1. An Input-restricted Deque
2. An Output-restricted Deque

An **Input-restricted deque** is a deque which allows insertions at only one end of the list but allows deletions at both ends of the list.

An **Output-restricted deque** is a deque which allows deletions at only one end of the list but allows insertions at both ends of the list.

If the element is added on the left, then LEFT is decreased by 1

If the element is added on the right, then RIGHT is increased by 1

If the element is deleted from the left, then LEFT is increased by 1

If the element is deleted from the right, then RIGHT is decreased by 1